

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
7	(a)	(i)		hard water contains calcium ions / Ca^{2+} / magnesium ions / Mg^{2+} (1) (Ca^{2+} / Mg^{2+}) swap places with sodium / Na^{+} ions (1) accept: swap $\equiv$ change $\equiv$ replaces $\equiv$ exchanges neutral answer: displaces reference to 'ions' needed only once	2			2		2
		(ii)		all Na^{+} ions have been removed / no more Na^{+} ions left (1) any soluble sodium salt e.g. sodium chloride (1) accept: salt solution	2			2		2
	(b)			add soap solution and shake (1) soft water forms lather (1) boil & add soap solution and shake / boil & repeat method (1) temporary hard water forms lather and permanent hard water doesn't (1)	4			4		4
				Question 7 total	8	0	0	8	0	8